

The-One-True-Archive: Series Index, Non-Redundancy Plan, and Cross-Pollination Map

Unified archive revision 2026-03-05 (v162 source-only)

1 What's new in v162 (contact-surface IDs + worked-example excerpt polish)

- **Synthesis 12:** added `contact_surface_id` to the minimal tuple schema table and clarified how to carry extra fields via `module_ext`; rebuilt PDF.
- **Synthesis 17:** expanded the receipt excerpt to show `fallback_contact_surface` with `contact_surface_id` + semantic interface fields; added an explicit digest-binding step (RFC 8785/JCS) to the client skeleton; rebuilt PDF.
- **Synthesis 30:** added a crisp rule-of-thumb separating long-lived deployment state (`state_decl_id`) from per-contact semantic interfaces (`contact_surface_id`); rebuilt PDF.
- rebuilt `SERIES_INDEX` and regenerated `MANIFEST.json`.

2 What's new in v161 (JCS digest binding + worked-example hygiene)

- **Synthesis 17:** fixed a stray `\noindent` typo, removed a lingering undocumented Rainbow protocol-filter knob mention, recommended RFC 8785 (JCS) for digest-bound canonical JSON, and labeled the shipped drift-case table; rebuilt PDF.
- **Synthesis 12:** standardized `contact_surface_id` as `sha256(JCS(contact_surface))` with an RFC 8785 anchor; rebuilt PDF.
- **Synthesis 30:** minor citation/title cleanup so the Rainbow changelog anchor no longer implies a protocol-filter knob; rebuilt PDF.
- rebuilt `SERIES_INDEX` and regenerated `MANIFEST.json`.

3 What's new in v160 (protocol-filter hygiene + worked-example drift cases + retrieval pinning)

- **Synthesis 30:** corrected protocol-filter discussion (removed undocumented Rainbow knob), and clarified that protocol-filter defaults must be pinned by deployed client version; rebuilt PDF.

- **Synthesis 12:** expanded service contact surfaces to include method and time budgets, and added a recommended `contact_surface_id` digest; rebuilt PDF.
- **Synthesis 17:** added canonicalization notes plus a 3-case drift taxonomy table and a `user_fast_diff` excerpt from the shipped compare report; rebuilt PDF.
- rebuilt `SERIES_INDEX` and regenerated `MANIFEST.json`.

4 What's new in v159 (routing budgets + retrieval surfaces + user micro-diff)

- **Synthesis 30:** added Rainbow's application-level routing time budget (`RAINBOW_ROUTING_TIMEOUT`) alongside network-level router timeouts, and documented Rainbow HTTP retrieval knobs (`enable/allow/deny/timeouts`) as a derived destination surface; rebuilt PDF.
- **Synthesis 12:** tightened service contact surfaces as semantic interfaces and expanded the state-surface description to include refresh/timeout budgets and fallback rules; rebuilt PDF.
- **Synthesis 17:** added a deterministic receipt-level micro-diff algorithm and generalized request-shaping drift to include routing/retrieval budgets; rebuilt PDF.
- **Synthesis 24:** refreshed archive stamp to v159; rebuilt PDF.

5 What's new in v158 (Rainbow autoconf refresh + line-item glance + attenuation intuition)

- **Synthesis 30:** made the “delegated router is a caching proxy” coupling explicit (IPNI docs) and added Rainbow control-plane knobs (`RAINBOW_AUTOCONF_REFRESH`, `RAINBOW_HTTP_ROUTERS_TIMEOUT`) to the deployed-surface checklist; rebuilt PDF.
- **Synthesis 17:** added an at-a-glance line-item table and three tiny attenuation intuition rows (keeps shipped bundle label `worked-example-draft-59`); rebuilt PDF.
- **Anonymity B:** refreshed archive stamp to v158; rebuilt PDF.
- regenerated `MANIFEST.json`.

6 What's new in v157 (protocol-filter surfaces + micro-diff hygiene)

- **Synthesis 30:** made delegated-routing *protocol filters* a first-class request-shaping surface across implementations by adding Rainbow's `--http-routers-filter-protocols` knob alongside Kubo's `IPFS_HTTP_ROUTERS_FILTER_PROTOCOLS`; updated the pinning checklist; rebuilt PDF.
- **Synthesis 17:** tightened the micro-diff discussion (single state-surface-drift rule, less redundancy) and fixed a stray tab/typo that caused an `exttt{...}` rendering glitch; rebuilt PDF.
- **Synthesis 12:** synchronized archive stamp to v157; rebuilt PDF.
- regenerated `MANIFEST.json`.

7 What's new in v156 (receipt-schema de-bloat + protocol-filter pinning)

- **Synthesis 12:** removed module-by-module prose (keeps only the stable tuple schema + drift semantics), and tightened the definition of contact surfaces to include path scope and request-shaping defaults; rebuilt PDF.
- **Synthesis 30:** made protocol filtering a first-class deployed-surface knob by pinning `IPFS_HTTP_ROUTERS_FILTER.F` and citing IPIP-0484 explicitly; rebuilt PDF.
- **Synthesis 17:** improved the worked example with a short client-side algorithmic skeleton and a concrete “request-shaping drift” micro-diff; kept shipped bundle label `worked-example-draft-59`; rebuilt PDF.
- regenerated `MANIFEST.json`.

8 What's new in v155 (worked-example tightening + override precedence)

- **Synthesis 17:** compressed the worked example wrapper (schema-nonredundant receipt slice + tiny excerpt), merged drift examples into the micro-diff section, and reduced the numerical slice to the single attenuation shape + validator pointer; rebuilt PDF.
- **Synthesis 30:** added an explicit “how to cite” ownership paragraph (defers receipt mechanics to Synthesis 12 and state change-control to Synthesis 18) and added a compact router-override precedence crosswalk tied to `state.decl_id`; rebuilt PDF.
- **Anonymity B:** added a remark coupling equalization targets to declared projections and pinned deployed state/control-plane surfaces; rebuilt PDF.

9 What's new in v154 (HTTP retrieval surfaces + env override pinning)

- **Synthesis 30:** refined AutoConf/override treatment: explicitly pinned `IPFS_HTTP_ROUTERS` and `IPFS_HTTP_ROUTERS_FILTER_PROTOCOLS` (IPIP-484), added Rainbow `RAINBOW_HTTP_ROUTERS` reference, and added a new short section on `HTTPRetrieval.Enabled` and trustless-HTTP block retrieval surfaces; rebuilt PDF.
- **Synthesis 12:** added a short paragraph noting that delegated-routing router sets and retrieval-mode switches are legitimate state line items owned by Synthesis 30; rebuilt PDF.
- **Synthesis 17:** tightened non-redundancy by pointing schema ownership to Synthesis 12 and control-plane drift ownership to Synthesis 30; kept shipped bundle label `worked-example-draft-59`; rebuilt PDF.
- **Synthesis 7:** upgraded Pisces citation from preprint-only to NDSS 2013 + canonical PDF; rebuilt PDF.

10 What

's new in v151 (worked-example line-item repair + deployed-surface tightening)

- **Worked-example line items (Synthesis 17):** actually repaired the missing row: `fallback_contact_surface` now appears in the “representative line items” table (matching the shipped receipt and the numerical slice).
- **Deployed surfaces note (Synthesis 30):** removed a duplicated gateway-drift paragraph, fixed a quotation glitch, and made the “self-hosted gateway as delegated router” surface explicit with the config knob `Gateway.ExposeRoutingAPI`; also added a short `cid.contact` “shared reader” sentence anchored to IPFS Docs.
- **Profiling contract stamp (Anonymity B):** refreshed archive stamp to v151 (content unchanged; pointer-chain still treats Synthesis 30 as the canonical deployed-surface map).

11 What

's new in v150 (citation closure + worked-example line-item coherence)

- **Deployed surfaces (Synthesis 30):** added a concrete drift surface from the current Kubo release line: `/routing/v1` is exposed on the gateway port by default in v0.40.0, making self-hosted gateways viable delegated-routing backends (privacy-relevant reader-set shift).
- **Threat windows (Synthesis 4):** added a pointer that delegated-routing/AutoConf/gateway defaults belong in the declared state surface; see Synthesis 30 for what-to-pin.
- **Related work map (Synthesis 7):** refreshed stamp to v149 and clarified the deployed-mitigations row to include self-hosted gateway exposure (keeps Synthesis 30 as the canonical deployed-surface map).

12 What's new in v147 (citation tightening + selection discipline bridge)

- **Deployed-surface note tightened (Synthesis 30):** anchored the public delegated-routing endpoint to IPFS Docs “Public IPFS Utilities” and replaced the weaker CLI anchor for `["auto"]` placeholder resolution with the Kubo RPC reference (`expand-auto`); bumped note to v0.04 and archive stamp to v147; rebuilt PDF.
- **Related-work map citation hygiene (Synthesis 7):** replaced a personal-hosted “open PDF” link for the SAC’25 Kademlia query-obfuscation paper with the TU Dresden research-portal record; bumped note to v0.44 and archive stamp to v147; rebuilt PDF.
- **Selection-discipline note made referee-friendlier (Synthesis 29):** updated archive stamp and added one external anchor to post-selection inference (Lee et al., Ann. Stat. 2016) to justify the pilot/frozen “data splitting” warning; rebuilt PDF.
- **Profiling contract stamp + citation hygiene (Anonymity B):** updated archive stamp and aligned the SAC’25 query-obfuscation bibliographic anchor with Synthesis 7; rebuilt PDF.

- **Worked-example clarity (Synthesis 17):** minor numerical-slice clarification (explicit tier contributions) without changing the shipped bundle label (`worked-example-draft-59`); archive stamp bumped to v147; rebuilt PDF.

13 What’s new in v146 (deployed-surface citation closure + worked-example stamp)

- **Synthesis 30 citation closure:** added an authoritative pointer to the canonical IPFS Mainnet AutoConf manifest (<https://conf.ipfs-mainnet.org/autoconf.json>), added a docs-based anchor for ‘auto’ placeholder resolution in Kubo, and replaced the chunked-OHTTP IESG approval placeholder with the IETF-Announce mailarchive record (27 Feb 2026); bumped note to v0.03 and archive stamp to v146; rebuilt PDF.
- **Worked-example stamp hygiene (Synthesis 17):** refreshed the archive stamp to v146 while keeping the shipped artifact bundle fixed at `worked-example-draft-59`; rebuilt PDF.

14 What’s new in v145 (deployed-surface accuracy hardening + pointer-chain coherence)

- **Deployed-surface note made referee-grade (Synthesis 30):** corrected and tightened the IPFS/libp2p control-plane citation stack (raw Kubo docs + IPFS Docs), clarified AutoConf defaults (conf.ipfs-mainnet.org/autoconf.json), added explicit pinning guidance for env-var overrides (`IPFS_HTTP_ROUTERS`), and clarified that the delegated-routing reader-privacy upgrade track (IPIP-0421) is currently a draft; bumped note to v0.02 and archive stamp to v145; rebuilt PDF.
- **Related-work map stays compact (Synthesis 7):** further de-bloated the deployed-mitigations branch to cite Synthesis 30 as the canonical deployed-surface map (keeping only measurement comparators inline); bumped note to v0.43 and archive stamp to v145; rebuilt PDF.
- **Worked-example pointer chain cleaned (Synthesis 17):** added an explicit pointer to Synthesis 30 for primary-path control-plane surfaces, repaired a missing `series:notation` bibliography anchor, and refreshed the archive stamp to v145 (`worked-example` bundle remains `worked-example-draft-59`); rebuilt PDF.
- **State series coherence (AnonDHT State 1):** refreshed stamp to v145 and added a one-line pointer to Synthesis 30 so deployed control-plane drift is treated as part of the declared state surface; rebuilt PDF.

15 What’s new in v144 (deployed-surface note + related-work de-bloat)

- **New deployed-surface companion note (Synthesis 30):** isolates IPFS/libp2p control-plane and service-concentration surfaces (delegated routing, AutoConf, OHTTP/chunking evolution) and provides a referee-grade “what-to-pin” checklist for published claims; built to reduce redundancy in Synthesis 7 and Anonymity B; rebuilt PDF.

- **Related-work map de-bloated (Synthesis 7):** moved detailed deployed-surface discussion into Synthesis 30, keeping Synthesis 7 as a compact taxonomy/index; bumped note to v0.42 and archive stamp to v144; rebuilt PDF.
- **Profiling paper pointer chain tightened (Anonymity B):** now cites Synthesis 30 at first mention of deployed control-plane surfaces; archive stamp bumped to v144; rebuilt PDF.
- **Index hygiene:** fixed a missing \item marker for Synthesis 29 and removed a duplicate Paper 12 inventory entry.

16 What’s new in v143 (deployed-surface citation hardening + selection-discipline pointers)

- **Deployed IPFS surfaces citation hardening (Synthesis 7):** keeps the control-plane/autoconf and caching-endpoint surfaces, but adds authoritative IPFS Docs and Kubo-config references for the public delegated-routing endpoint and AutoConf configuration knobs; bumped note to v0.41 and archive stamp to v143; rebuilt PDF.
- **Profiling contract hygiene (Anonymity B):** repaired an archive-stamp drift (v143), replaced a placeholder wiki-style Prefix Capacities entry with the series’ standard published-note pointer, and added an explicit selection-discipline reminder in the log-audit workflow when projections use data-selected bucket/coarsening maps; rebuilt PDF.
- **Worked-example non-redundancy (Synthesis 17):** coarsening-variant paragraph now explicitly imports selection discipline (Synthesis 29) for any data-selected partition maps; archive stamp bumped to v143; rebuilt PDF.

17 What’s new in v142 (worked-example coherence + deployed autoconf surface)

- **Worked example coherence fix (Synthesis 17):** corrected an internal drift: the narrative stamp now matches the shipped worked-example artifacts (`note_version=0.59`, `worked-example-draft-59`) so auditors/users are not asked to reconcile mismatched ids; added one paragraph clarifying that primary-path protocol/endpoint specifics live in the manifest’s evidence hooks rather than bloating the public guard object; rebuilt PDF.
- **Deployed delegated-routing surfaces sharpened (Synthesis 7):** adds dynamic endpoint discovery / autoconf as an explicit control-plane observation surface (with a stable pointer to IPFS Mainnet AutoConf); bumped note to v0.40 and archive stamp to v142; rebuilt PDF.
- **Citation hardening (Anonymity B):** added DOI anchors for Peer2PIR (IEEE S&P’25) and Backes et al. (ASIACCS’12) to reduce “preprint-only” ambiguity; rebuilt PDF.
- **Stamp hygiene (AnonDHT State 1):** aligned archive stamp to v142; rebuilt PDF.

18 What’s new in v141 (citation closure + deployed-surface pointers)

- **Citation closure:** repaired dangling `series:selection` citations by standardizing on `series:selectiondiscipline` (Anonymity B and Synthesis 28); rebuilt PDFs.
- **Synthesis 28 refreshed:** stamp bumped to archive v141 and the selection-discipline note is now a first-class bibliography anchor (Synthesis 29); rebuilt PDF.
- **Related-work map sharpened (Synthesis 7):** adds explicit pointers to delegated-routing caching and Someguy-backed public endpoints as real deployed observation surfaces; stamp bumped to archive v141; rebuilt PDF.

19 What’s new in v140 (coherence + citation hardening)

- **Synthesis 7 (related work map) refreshed:** fixed a DOI typo for the 2025 Kademlia query-obfuscation paper and updated the archive stamp.
- **Synthesis 8 (notation/model) refreshed:** now explicitly points claims with data-dependent bucket maps to the selection-discipline note (Synthesis 29).
- **Synthesis 17 (worked example) tightened again:** adds an explicit import of the series-wide notation (Synthesis 8). (A later correction in v142 repairs a stamp drift so the paper matches the shipped worked-example artifacts.)
- **State 1 citation hardening:** replaces the Nym-reputation pointer with the ePrint report (2026/101) as a stable, referee-grade anchor.
- **Anonymity B (profiling) workflow note:** ties log-based equalization audits to selection-discipline when the audited projection uses learned/coarsened buckets.

20 What’s new in v138 (selection discipline for data-dependent maps)

- **New selection discipline note (Synthesis 29):** added a conservative, receipt-grade way to handle data-dependent bucket maps and coarsening partitions, centered on witness-indexed families and pilot/frozen phases under change control.
- **Non-redundancy tightened:** Synthesis 27 (bucket design) and Synthesis 28 (coarsening) now cite Synthesis 29 instead of re-explaining pilot/freeze and top- m selection caveats inline.
- **Profiling contract imports extended (Anonymity B):** adds a single pointer to Synthesis 29 so deployers see where data-dependent map selection belongs in the series stack.

21 What’s new in v137 (worked example now covers coarsening / large-alphabet audits)

- **Worked example expanded (Synthesis 17):** bumped to worked-example-draft-59 and added a fully wired *coarsening* variant receipt (large-alphabet contact-set hash projected via a digest-

bound deterministic partition map), including new artifacts, replay plan, and support-bundle resolver entries.

- **Coarsening note grounded (Synthesis 28):** now points explicitly to the Synthesis 17 coarsening variant so the design rule has a concrete receipt/evidence instantiation.
- **Profiling contract pointer chain tightened (Anonymity B):** adds an explicit pointer to the coarsening variant receipt so implementers see one end-to-end example without re-deriving audit feasibility.
- **Spectral delegation note refreshed (Anonymity D):** stamp aligned to archive v137 and the MaxL-to-odds-inflation bridge is made explicit via Synthesis 5.

22 What’s new in v136 (coarsening rules for large-alphabet audits)

- **New coarsening note (Synthesis 28):** added a receipt-facing, data-processing-safe way to make high-cardinality interface projections sample-feasible for log audits by declaring an explicit coarsening/partition map.
- **Bucket-budget lemma tightened (Synthesis 27):** replaced the linear-only rule of thumb with the exact alphabet constraint (and retained the approximation as a secondary line) so auditors can see the regime where certification is impossible.
- **Profiling contract pointer chain improved (Anonymity B):** imports coarsening guidance from Synthesis 28 and cites it in the “value of coarsening” research agenda, reducing the temptation to re-derive sampling arguments in mechanism/contract papers.

23 What’s new in v135 (bucket design for continuous timing projections)

- **New timing-bucket note (Synthesis 27):** added a compact design and publication checklist for turning continuous timing telemetry into a declared *discrete* projection suitable for receipt-grade equalization claims (and for Synthesis 25–26 log-audit certificates).
- **Non-redundancy tightened:** Synthesis 25 now points to Synthesis 27 for bucket-map design rather than re-explaining timing discretization; Anonymity B similarly cites Synthesis 27 at first mention of timing buckets.
- **Index hygiene:** the inventory and reading-order map now list Synthesis 27 explicitly.

24 What’s new in v133 (menu-level log-audit note + de-duplication)

- **New statistical audit note (Synthesis 25):** added a short, conservative menu-level finite-sample TV certificate for discrete interface projections audited from logs, with a receipt-facing publication checklist and an explicit many-testing correction parameter δ_{stat} .
- **Profiling contract tightened (Anonymity B):** removed an inlined Clopper–Pearson audit lemma (now centralized in Synthesis 25) so the paper stays mechanism/contract focused; added a clean pointer chain to the worked-example receipts (Synthesis 17).

- **ABOM/OINL hygiene (Anonymity C):** made the ABOM definition explicitly defer to the shared contract-object vocabulary (Synthesis 0) and updated the minimal schema sketch to avoid stale path references; stamps aligned to archive v133.
- **Rebuild + manifest:** rebuilt touched PDFs, updated `PATCH_NOTES.md` and regenerated `MANIFEST.json`.

25 What’s new in v126 (tight non-redundancy + restricted-route branch + evidence drift)

- **Related work map (Synthesis 7):** added a restricted-route / social-overlay DHT routing branch (X-Vine, R⁵N) to prevent conflating “anonymous lookup” with “directory scaling”; added compact citations to the academic papers and the R⁵N technical specification.
- **Worked example (Synthesis 17):** upgraded to worked-example-draft-55 (v0.55) and replaced the third drift vignette with an *evidence drift* example centered on `evidence_id` + `replay_hook` (projection/audit knob changes are diffable).
- **Profiling contract (Anonymity B):** clarified that adaptive stopping/drift monitoring should use the state-series anytime-valid (e-value) audits rather than the i.i.d. log-certificate lemma.

26 What’s new in v125 (receipt completeness + stopping-safe drift audits + rebuilds)

- **Packaging correctness:** rebuilt missing PDFs (Operational A and Synthesis 12) and recompiled touched notes.
- **State-dependent anonymity (AnonDHT State 1):** added a stopping-safe drift-audit subsection based on e-values / Ville’s inequality (time-uniform validity under optional stopping), and removed stale inlining markers.
- **Receipt schema clarity (Synthesis 12):** added an explicit remark separating `evidence_id` (supporting object) from `replay_hook` (recomputation plan).
- **TeX hygiene:** removed legacy “inlined section” comment scaffolding across multiple papers to keep sources tight and readable.
- **Index + manifest:** updated this index, patched notes, and regenerated `MANIFEST.json`.

27 What’s new in v123 (conditional certificates fixed + worked-example de-bloat + OHTTP draft refresh)

- **Conditional per-step certificates repaired (Synthesis 21):** fixed a real TeX corruption so the certificate-shape note compiles cleanly (and is again safe to cite as the Regime II interface); rebuilt PDF.
- **Worked example tightened without losing coverage (Synthesis 17):** upgraded to worked-example-draft-54 (v0.54), replaced the long object inventory table with a grouped pointer to the shipped artifact inventory + support manifest, regenerated artifacts, and rebuilt PDF.

- **Related-work map refreshed (Synthesis 7):** updated the chunked OHTTP anchor to draft-ietf-ohai-chunked-ohttp-08 (18 Feb 2026) and rebuilt PDF.
- **Receipt schema note clarified (Synthesis 12):** `state_decl_id` is explicitly optional (stateful/monitored settings) and `witness_tier` is non-binding; rebuilt PDF.
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated MANIFEST.json.

28 What’s new in v122 (CP+union-bound log audit; computed profiling evidence)

- **Profiling audit made referee-grade (Anonymity B):** separated the statistical confidence parameter δ_{stat} from the published slack δ , and used exact binomial (Clopper–Pearson) intervals with a union bound to compute a deterministic receipt-grade (η, δ) summary from a published count table; rebuilt PDF.
- **Worked example reconciled and computed (Synthesis 17):** fixed a mismatch between the narrative slice and the receipt’s profiling values, upgraded to worked-example-draft-53 (v0.53), and updated the generator so the profiling (η, δ) is computed from toy counts via the same CP+union-bound procedure (evidence schema v2); rebuilt PDF.
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated MANIFEST.json.

29 What’s new in v120 (notation refresh + worked example now covers profiling certificates)

- **Notation sheet upgraded (Synthesis 8):** clarified line-item interface IDs (`exposure_nf_id`, `cond_iface_id`) and introduced an explicit `evidence_id` convention so log-audits and per-step certificates can be linked without repeating ID semantics across papers.
- **Worked example broadened without bloat (Synthesis 17):** added a fourth representative line item for profiling/equalization ((
eta,
delta)) to demonstrate that log-audit evidence objects plug into the same receipt/replay surface as routing-path claims.
- **Related work map stamp refreshed (Synthesis 7):** bumped the archive stamp to v120 (no taxonomy changes).
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated MANIFEST.json.

30 What’s new in v119 (profiling audit now yields (η, δ) + horizon/certificate interlock)

- **Profiling audit made receipt-grade (Anonymity B):** replaced the finite-sample audit lemma with an explicit (η, δ) *equalization certificate* for discrete projections. The slack δ now accounts for rare bins directly (no ad-hoc probability floor), and the paper states the coarsening rule needed to drive δ small.

- **Horizon claims tied to receipt fields (Synthesis 19):** added a one-paragraph mapping from “horizon- n ” prose to the actual line-item hooks (`tw_id`, `usage`, `replay_hook`) and pointed Regime II authors to the conditional-certificate interface.
- **Per-step certificates given a minimal record format (Synthesis 21):** added a compact `{step_selector, cond_iface_id, bound_bits, evidence_id}` shape for plan-spec tooling output, with the profiling/equalization audit as a canonical log-audit evidence object.
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated `MANIFEST.json`.

31 What’s new in v118 (worked-example de-bloat + profiling audit lemma + OHTTP discovery pin)

- **Worked example tightened (Synthesis 17):** removed schema restatements and large redundant interlock tables; the note now exercises only three representative line items and points to the owning papers for schemas and theory.
- **Profiling audit made more operational (Anonymity B):** added a conservative finite-sample max-divergence audit lemma for discrete transcript projections (log-based certification without bespoke tooling).
- **Related-work map updated (Synthesis 7):** added RFC 9540 (OHTTP service discovery via SVCB/HTTPS) as an explicit “metadata surface to pin” alongside RFC 9458 and chunked OHTTP drafts.
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated `MANIFEST.json`.

32 What’s new in v117 (related-work map expansion + PDF build fix)

- **Related-work map expanded (Synthesis 7):** added an explicit “DHT-backed relay-selection/directory” branch (AP3, NISAN, Torsk) to separate classic DHT-privacy mechanisms from Tor-style directory scaling, added a Bitswap random-walk plausible-deniability anchor, and added a chunked-OHTTP draft anchor (metadata corrected in v122).
- **Packaging correctness:** regenerated previously-empty PDFs by recompiling their `paper.tex` sources (no interface changes implied).
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated `MANIFEST.json`.

33 What’s new in v116 (worked-example tightening + TW/ENF interlocks)

- **Worked example upgraded (Synthesis 17):** added an explicit “guarded endpoint” form that carries the separation-failure bound ρ alongside the scalar MaxL summary, so the public receipt no longer silently drops the primary-path guard. Also added a short math-backbone pointer paragraph so the example can stay non-redundant.

- **Trace/Threat notes aligned to receipts (Synthesis 3–4):** tightened the downstream pointer to `exposure_nf_id` and `tw_id` usage in line items, and added a concrete TW-template instance used by the worked example.
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated `MANIFEST.json`.

34 What’s new in v115 (previous revision)

- **Separation manifests made more replayable (Synthesis 22):** the minimal manifest shape now commits the *assignment/rotation policy* (how relays/routers are selected and rotated), which is the missing hook needed to justify any published separation-failure bound.
- **Probabilistic separation now has a model-to- ρ crosswalk (Synthesis 23):** added a short set of “toy bounds” that translate common committed clauses (independent compromise; finite-adversary counts; per-epoch rotation) into a window-level ρ .
- **Related work updated for evolving OHTTP transports (Synthesis 7):** added the OHAI chunked-OHTTP draft and a short warning that chunking introduces new metadata features that must be pinned under the declared projection.
- **Rebuild + manifest:** rebuilt touched PDFs and regenerated `MANIFEST.json`.

35 Backbone papers (published; canonical wiki links)

These five are treated as the *stable* references that later papers cite instead of repeating core machinery: The scheduling/compiler note is materially longer than the other backbone papers; on a first pass, many downstream papers need only its stop-time-padding, public-schedule, and TV-certificate interfaces, with the full optimization stack available as a deeper reference.

- [[2026.01.22 - Mathematics: Scheduling and Compiler Techniques for Metadata-Hiding Overlay Lookups]]
- [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]]
- [[2026.01.25 - Mathematics: Prefix Capacities and Separation Cuts for Shared-Kernel Termination-Time Hiding]]
- [[2026.01.26 - Mathematics: Stop-Time Padding Addendum, Max-Mixture Separations and Tail-Sign Deadline Normal Forms]]
- [[2026.01.29 - Mathematics: Optimal Poset-Feasible Padding: Knapsack, Transport, and Dual Certificates for TV Privacy]]

36 Archive invariants (stable policy)

- **Single-file rule:** each paper is one `paper.tex` (no input splits, no external `.bib`).
- **No hidden build flags:** extended material is shipped as explicit companion addendum papers (still single-file), not gated behind `\iflong` toggles.

- **Source-only packaging:** figures are replaced by boxed placeholders; plots/scripts/datasets are available on request (see Synthesis 6 for the artifact policy).
- **Owned background:** repeated background is minimized by assigning ownership to specific papers/series and citing them elsewhere.
- **Stable outward links:** cross-paper pointers are normalized to the canonical published wiki links (Backbone section above).
- **Patch notes + file manifest:** the latest narrative delta lives in `PATCH_NOTES.md`; source-file digests live in `MANIFEST.json`.

37 Continuity map from older split archives

Older split archives bundled some “supplement” notes and long-build sweeps for operator-facing microstudies. In this unified archive, we preserve that material *without* hidden compile toggles by using an explicit “core paper + addendum” pattern: the main paper stays referee-readable and citable; the addendum carries supporting tables, microstudies, and longer sweeps.

- **PSC-Q split supplement** → `congestion_series/paper2A_psc_q_microstudies_addendum/paper.tex`.
- **Boss Fight dial-sheet sweeps** → `bossfight_series/paperB_addendum_evidence_tables/paper.tex`.
- **Evaluation 1 long figures/templates** → `evaluation_series/paper1A_notarized_certificates_addendum/`
- **State-series calibration/monitoring extensions** → `anondht_state_series/paper1A_...` and `paper3A_...` addenda.
- **Congestion-EQ / W-Congestion-EQ legacy microstudies** remain summarized inline (plots/scripts on request) because their most salient content is conceptual rather than tabular.

This keeps the non-redundancy contract intact while ensuring a default compile includes the core argument and all companion notes remain independently checkable.

38 Non-redundancy policy (gameplan)

To keep papers brief while referee-grade:

1. **Each endpoint metric is “owned” once.** Odds inflation / composition is owned by *Odds-Inflation Anonymity* (anonymity series Paper A). TV certificates and duality are owned by the published knapsack/transport paper. Congestion witness definitions are owned by *Congestion-EQ*.
2. **Each new witness tier must export a contract object.** If a paper introduces retries, calendars, state evolution, or auditing, it should define (i) the witness family, (ii) the budget metric, and (iii) the accountant for repeats/resets, then cite prior papers for everything else.
3. **Certification is a separate layer.** Mechanism papers should avoid re-litigating certificate formats. ABOM/OINL, proof-carrying budgets, notarization, and ledger receipts are the canonical certification papers.
4. **No duplicated “background” sections.** When background is required, use a single paragraph pointer to the backbone paper and restate only the local notation.

39 Cross-pollination targets (high leverage)

This is the shortlist of “shared lemmas” / interfaces worth reusing across papers to prevent drift:

1. **Trace interfaces / tiers:** name the transcript projection first. Reuse the CCT schema and tier vocabulary (Synthesis 3) whenever a paper budgets “the transcript”.
2. **Endpoint metrics bridge:** reuse the equivalences among realized odds inflation, PML, and MaxL, and the standard $\log(\alpha K)$ identification sizing rule (Synthesis 5) rather than re-deriving them locally.
3. **Spectral vs. periodic mechanisms:** spectral delegation bounds can be reused to reason about periodic calendars and declared dither (PSC-Q).
4. **Retries everywhere:** retries are an attack surface in lookups *and* in evaluation pipelines. Reuse the notarization / public randomness pattern for both runtime and evaluation.
5. **State privacy $\leftrightarrow$ monitoring:** state-dependent anonymity turns caches/scores into witnesses; closed-view auditing turns witnesses into receipts. The unifying principle is “account for what you publish, and publish what you account.”
6. **Release receipts $\leftrightarrow$ anonymity certificates:** both are transparency-log objects with an equivocation threat model. Reuse the same log schema and auditor tooling where possible.

40 Congestion series capsule (preserved from the old split index)

The congestion sub-series is intentionally a three-paper pipeline:

- **Congestion-EQ** defines the *congestion witness* and a TV-style budget/accountant for RTT, queueing delay, and loss/ECN marks.
- **PSC-Q** provides *mechanisms* (public service calendars and substitution padding) that meet Congestion-EQ budgets in a compiler-friendly way.
- **W-Congestion-EQ** defines *verifiability*: certificates, verifiers, and transparency-log anchoring for published congestion-budget claims.

41 Reading order (suggested)

1. Backbone papers (five wiki links above).
2. Synthesis 8 (notation/model) as a short shared preface for the rest of the archive; Synthesis 24 as a compact crosswalk from the five backbone math notes to the receipt-facing interfaces; and Synthesis 25–26 as the centralized log-audit layer (discrete finite-sample certificates + anytime spending/change control).
3. Anonymity series: Paper A (odds inflation) $\rightarrow$ Paper B (profiling) $\rightarrow$ Paper C (ABOM/OINL) $\rightarrow$ Paper D (spectral delegation certificates).
4. Boss Fight series: budgets $\rightarrow$ dial-sheet $\rightarrow$ proof-carrying budgets.

5. Optional early detour for architectural motivation: AnonDHT State 2 (MUCC / committee-contact privacy), because it gives the clearest “why not just equalize the raw DHT everywhere?” impossibility result in the archive.
6. Congestion series: Congestion-EQ → PSC-Q → W-Congestion-EQ.
7. AnonDHT state series (optional but important for modern systems): state-dependent anonymity → MUCC → closed-view auditing. Read MUCC earlier if you want the strongest motivation for the primary/fallback architectural pattern.
8. Evaluation series and release receipts (when publishing claims publicly under change control).
9. Series synthesis notes, with paper-type cues for the reader: Synthesis 8 (notation/model note) → Synthesis 24 (math-backbone crosswalk note) → Synthesis 27 (timing bucket design note) → Synthesis 28 (coarsening/large-alphabet note) → Synthesis 29 (selection-discipline note) → Synthesis 25 (log-audit certificate note) → Synthesis 26 (anytime spending/change control note) → Synthesis 0 (contract-object interface note) → Synthesis 3 (trace/schema note) → Synthesis 4 (threat-window checklist note) → Synthesis 19 (receipt-accounting rulebook note) → Synthesis 5 (metric-bridge note) → Synthesis 12 (receipt-schema note) → Synthesis 18 (plan/state equivalence note) → Synthesis 20 (auditor replay checklist note) → Synthesis 21 (Regime II certificate interface note) → Synthesis 22 (separation-manifest guard note) → Synthesis 23 (probabilistic separation adjunct note) → Synthesis 13 (user-discovery/validation note) → Synthesis 17 (worked-example/interlock micro-note) → Synthesis 10 (architecture note) → Synthesis 11 (math/interface note on attenuation) → Synthesis 14 (math note on path-selection tax) → Synthesis 15 (receipt-composition note) → Synthesis 9 (transparency interface note) → Synthesis 7 (related-work map) → Synthesis 30 (deployed-surfaces companion note) → Synthesis 6 (policy micro-note). If you want to see the public objects before the architecture story, read Synthesis 17 immediately after Synthesis 12–13; it is the archive’s concrete pressure test for receipt publication, UVI discovery, and evaluation handoff.

42 Paper inventory (paths in this archive)

Each entry is a single-file `.tex` paper.

- `published/2026-01-22_scheduling_compiler/paper.tex`
- `published/2026-01-23_spectral_anonymity/paper.tex`
- `published/2026-01-25_prefix_capacities/paper.tex`
- `published/2026-01-26_stop_time_padding_addendum/paper.tex`
- `published/2026-01-29_knapsack_transport/paper.tex`
- `published/optional_many_testing_lower_bounds/paper.tex` (optional deep-dive)
- `anonymity_series/paperA_odds_inflation_anonymity/paper.tex`
- `anonymity_series/paperB_anonymous_dht_profiling/paper.tex`
- `anonymity_series/paperC_abom_oiml/paper.tex`
- `anonymity_series/paperD_spectral_delegation_certificates/paper.tex`

- bossfight_series/paperA_bossfight_budgets/paper.tex
- bossfight_series/paperB_anondht_dial_sheet/paper.tex
- bossfight_series/paperB_addendum_evidence_tables/paper.tex
- bossfight_series/paperC_proof_carrying_budgets/paper.tex
- certified_series/paperA_certified_menus_anonymous_dht/paper.tex
- certified_series/paperB_routing_signature_compression/paper.tex
- certified_series/paperC_disagreement_gated_recertification/paper.tex
- congestion_series/paper1_congestion_eq/paper.tex
- congestion_series/paper2_psc_q/paper.tex
- congestion_series/paper2A_psc_q_microstudies_addendum/paper.tex
- congestion_series/paper3_w_congestion_eq/paper.tex
- anondht_state_series/paper1_state_dependent_anonymity/paper.tex
- anondht_state_series/paper1A_state_dependent_anonymity_calibration_addendum/paper.tex
- anondht_state_series/paper2_mucc_committee_contact_privacy/paper.tex
- anondht_state_series/paper3_closed_view_auditing_cppc/paper.tex
- anondht_state_series/paper3A_cppc_addendum/paper.tex
- operational_series/paperA_schedule_only_retries_anondht/paper.tex
- operational_series/paperB_auditable_trace_privacy/paper.tex
- evaluation_series/paper1_notarized_anonymity_certificates/paper.tex
- evaluation_series/paper1A_notarized_certificates_addendum/paper.tex
- evaluation_series/paper2_advantage_contracts_stealth_audits/paper.tex
- evaluation_series/paper3_calibration_recipes_anondht/paper.tex
- release_and_destination/paperA_release_property_ledger_receipts/paper.tex
- release_and_destination/paperB_mceq_coversketch_destination_privacy/paper.tex
- synthesis/paper0_contract_objects/paper.tex
- synthesis/paper1_receipt_redaction_bounds/paper.tex
- synthesis/paper2_calendar_synthesis_duals/paper.tex
- synthesis/paper3_trace_interfaces_dht/paper.tex
- synthesis/paper4_threat_windows_anondht/paper.tex

- `synthesis/paper5_endpoint_metrics_bridge/paper.tex`
- `synthesis/paper6_artifact_policy/paper.tex`
- `synthesis/paper7_related_work_map/paper.tex`
- `synthesis/paper30_deployed_surfaces_control_plane/paper.tex`
- `synthesis/paper8_notation_model/paper.tex`
- `synthesis/paper24_math_backbone_crosswalk/paper.tex`
- `synthesis/paper25_discrete_equalization_audits/paper.tex`
- `synthesis/paper26_anytime_menu_spending/paper.tex`
- `synthesis/paper27_bucket_design_continuous_timing/paper.tex`
- `synthesis/paper28_coarsening_large_alphabet/paper.tex`
- `synthesis/paper29_selection_discipline_datadependent/paper.tex`
- `synthesis/paper9_transparency_primitives_interface/paper.tex`
- `synthesis/paper10_tiered_architecture_primary_fallback/paper.tex`
- `synthesis/paper11_observation_attenuation/paper.tex`
- `synthesis/paper12_receipt_line_items_schema/paper.tex`
- `synthesis/paper18_plan_and_state_equivalence/paper.tex`
- `synthesis/paper13_user_verifiability_minimal_interface/paper.tex`
- `synthesis/paper14_fallback_selection_tax/paper.tex`
- `synthesis/paper15_tiered_observation_vectors/paper.tex`
- `synthesis/paper19_receipt_accounting_rulebook/paper.tex`
- `synthesis/paper20_auditor_replay_recipe/paper.tex`
- `synthesis/paper21_conditional_budget_certificates/paper.tex`
- `synthesis/paper22_primary_separation_manifests/paper.tex`
- `synthesis/paper23_probabilistic_separation_budgets/paper.tex`
- `synthesis/paper16_deployment_blueprint_primary_fallback/paper.tex`
- `synthesis/paper17_worked_example_receipt_interlock/paper.tex`
- `congestion_series/series_index/paper.tex`